Module 9: Inheritance Genetics — Study Questions

1. Who is Gregor Mendel? What organism did he study?
2. What is the difference between a dominant allele and a recessive allele?
3. What is the difference between genotype and phenotype?
4. What is the difference between homozygous (AA, aa) and heterozygous (Aa)?
5. Cross Tt × Tt using a Punnett square. What is the phenotypic ratio?
6. What is a test cross? What is it used for?
7. What phenotypic ratio do you expect from a dihybrid cross (AaBb × AaBb)?
8. What is incomplete dominance? Give an example.
9. What is codominance? Give an example.
10. Explain the ABO blood type system. What are the possible genotypes for Type A blood?
11. Can two parents with Type A and Type B blood have a Type O child? Explain.
12. Why are X-linked recessive disorders more common in males?
13. A carrier mother (X^B X^b) and a normal father (X^B Y) have children. What fraction of sons will be colorblind?
14. Two unaffected parents have an affected child. Is the trait dominant or recessive?
15. What is a pedigree? What can it tell you?
16. What is a polygenic trait? Give one example.
17. How is the inheritance of human height different from the inheritance of pea plant height?

18. True or False: If you know someone's genotype, you always know their phenotype. Explain.
19. What is Mendel's Law of Segregation? How does it connect to meiosis?
20. What is Mendel's Law of Independent Assortment?